

Zihan Liang

PHD STUDENT · COMPUTATIONAL BIOLOGY AND BIOINFORMATICS PROGRAM, DUKE

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Education

Duke University

Durham, NC

PH.D. IN COMPUTATIONAL BIOLOGY AND BIOINFORMATICS

Aug. 2026 - present

- Ph.D. Advisor: [Dr. Anru Zhang](#)

Emory University

Atlanta, GA

BACHELOR OF SCIENCE IN APPLIED MATHEMATICS AND STATISTICS WITH HIGHEST HONORS

Aug. 2023 - May 2026

- Minors in Artificial Intelligence
- Independent Research and Honors Program Advisor: [Dr. Ruoxuan Xiong](#)
- Relevant Courseworks:
 - Mathematics: Multivariable Calculus, Linear Algebra, Discrete Mathematics, Differential Equations, Linear Optimization, Graph Theory (Graduate Level), Real Analysis
 - Statistics & Computation: Mathematical Statistics, Numerical Analysis, Data Structures & Algorithms, Data Science Computing, Machine Learning, Text Data Science, Information Visualization, Experimental Methods, Causal Inference (Doctoral Level)

Publications

PUBLISHED

- Pan, Z^{*}, **Liang, Z^{*}**, Xiong, R. 2026. Two-Sided State-Space Models for Sequential Recommendation with Non-Random Multimodal Review Feedback. Findings of the Association for Computational Linguistics: EMNLP 2026.
- Pan, Z^{*}, **Liang, Z^{*}**, Kabbara, J., Emami, A. 2026. DART: Mitigating Harm Drift in Difference-Aware LLMs via Distill-Audit-Repair Training. Findings of the Association for Computational Linguistics: ACL 2026.
- Liang, Z^{*}**, Pan, Z^{*}, Xiong, R. 2026. Learning Dynamic Representations and Policies from Multimodal Clinical Time-Series with Informative Missingness. Findings of the Association for Computational Linguistics: ACL 2026.
- Shen, Y^{*}, Jiang, D^{*}, **Liang, Z^{*}**, Wei, Y^{*}, Yang, W., Sun, X., Huang, Z., Wang, S., Yao, Q. 2026. MambaDATG: Domain-Adaptive Tri-Plane-Gated Pre-training for 3D Abdominal Segmentation. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2026), Oral Presentation.
- Liang, Z^{*}**, Pan, Z^{*}, Xiong, R. 2025. Causal Representation Learning from Multimodal Clinical Records under Non-Random Modality Missingness. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025).
- Liang, Z^{*}**, Pan, Z^{*}, Dey, S. K., Ismail, A. 2025. CareLab at #SMM4H-HearD 2025: Insomnia Detection and Food Safety Event Extraction with Domain-Aware Transformers. Proceedings of the International AAAI Conference on Web and Social Media (ICWSM 2025) Workshops.
- Liang, Z.**, Pan, Z. 2026. Dynamic Policy Design for Autonomous Taxi Adoption: A Hierarchical Game-Theoretic Framework. Transportation Research Board 105th Annual Meeting (TRB 2026).

IN REVIEW

- Liang, Z^{*}**, Pan, Z^{*}, Yu, S. 2026. Adaptive Spatiotemporal Graph Neural Networks with Trend-Aware Prediction and Validation-Gated Calibration for Traffic Flow Forecasting. IEEE Transactions on Intelligent Transportation Systems, under review.

Research Experience

Collective Action & Research for Equity (CARE) Lab, Emory University

Atlanta, GA

RESEARCH ASSISTANT, ADVISOR: DR. AZRA ISMAIL

Sept. 2024 - May 2026

- Built an interactive dashboard for nearly 1,000 user records using JavaScript, Python, and HTML.
- Maintained a WhatsApp education bot serving 15,000+ teachers and 600,000+ students across India.
- Developed an offline-first Android app for classroom data collection and secure Google Drive synchronization.

Polymath Jr REU Summer Program

Remote

RESEARCH ASSISTANT

Jun. 2025 - Aug. 2025

- Developed ODE-based models to study community transmission of *C. difficile*.
- Evaluated intervention strategies for controlling community spread of *C. difficile*.
- Implemented disease simulations and mitigation analyses in MATLAB and R.

Honors & Awards

2026 Elected to Phi Beta Kappa (ΦBK), Phi Beta Kappa Society

2025 Best Visualization Award, American Statistical Association DataFest

2025 Conference Presentation Grant, Emory University

\$2,500

2023, 2024 Dean's List, Emory College of Arts and Sciences

2022 American Invitational Mathematics Examination (AIME), Score: 9/15, Mathematical Association of America

Teaching Experience

2026 Teaching Assistant and Grader, Department of Data and Decision Sciences, Emory University

2025 Teaching Assistant and Grader, AwesomeMath - Algebra Course for Mathematical Olympiads

Outreach & Professional Development

SERVICE AND OUTREACH

2023-2026 Emory Chinese Student Association, President

Atlanta, GA

2024-2026 Emory Artificial Intelligence & Data Association, Founder and President

Atlanta, GA

2022-2024 YouthBuild Program - Xi'an, Co-Founder & Standing Council Member

Xi'an, Shaanxi

PEER REVIEW

Reviewer, ACL 2026 Workshop on Social Media Mining for Health and Health Real-World Data (SMM4H-HeaRD)

Reviewer, ACM CIKM 2026 Workshop on LLM Agents for Social Simulation (LASS)